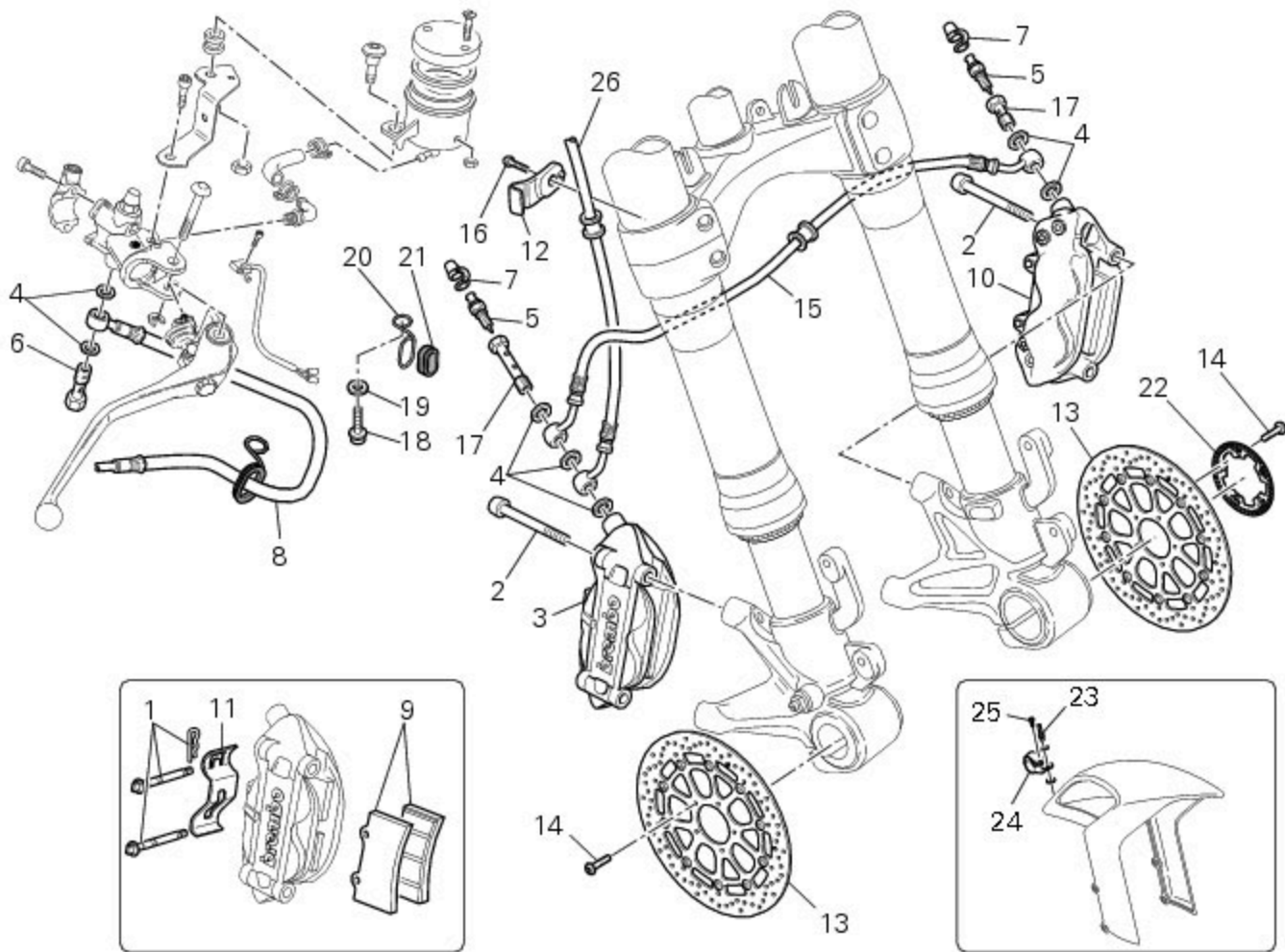


3 -Hydraulic front brake



- 1 Pad retaining pin and clip
- 2 Front brake calliper screw
- 3 Right brake calliper
- 4 Sealing washer
- 5 Bleed valve
- 6 Banjo bolt
- 7 Dust cap
- 8 Front pump - control unit pipe
- 9 Pair of brake pads
- 10 Left brake calliper
- 11 Springs
- 12 Bracket
- 13 Brake disc
- 14 Brake disc screw
- 15 Front brake hose
- 16 Screw
- 17 Banjo bolt
- 18 Screw
- 19 Washer
- 20 Ring
- 21 Rubber pad
- 22 Phonic wheel
- 23 Rivet
- 24 Hose clip
- 25 Hose clip screw
- 26 Control unit - front calliper pipe

1100 ABS [FRONT AND REAR WHEELS](#)

1100 ABS [FRONT BRAKE](#)

1100 S ABS [FRONT AND REAR WHEELS](#)

1100 S ABS [FRONT BRAKE](#)

Important

Bold reference numbers in this section identify parts not shown in the figures alongside the text, but which can be found in the exploded view diagram.

Maintenance operations

Warning

Brake fluid is corrosive and will damage paintwork. Avoid contact with eyes and skin. In case of accidental contact, wash the affected area with abundant running water and consult a doctor if necessary.

For all the maintenance operations (wear check and replacement of brake pads and brake fluids, etc.) see Sect. D 4, [Draining the brake circuit](#).

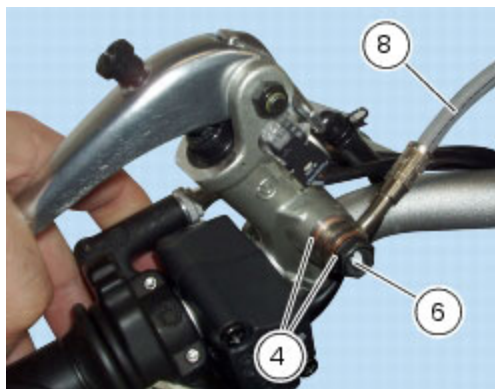
Removal of the front brake system

Operation	Section reference
Draining the braking circuit	D 4, Draining the brake circuit

Note

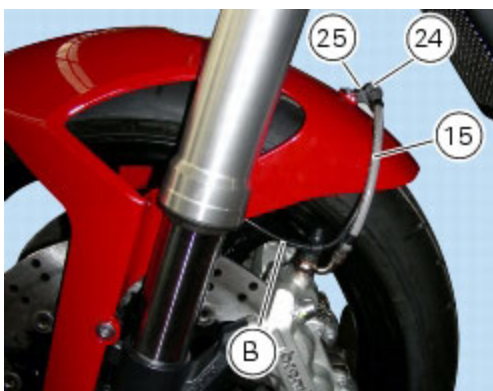
For the ABS front braking system, also refer to Sect. G 9.1, [ABS system operating information](#), Sect. G 9.2, [System components](#), Sect. G 9.3, [ABS components maintenance](#).

Unscrew the banjo bolt (6) securing the brake hose to the master cylinder and recover the sealing washers (4); slide hose (8) out of front master cylinder.

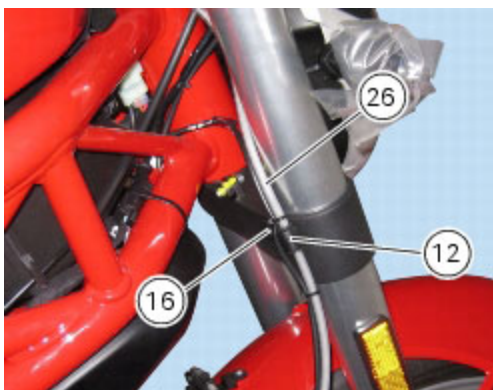


Slide the pipe (15) out of clips (A) and the pipe (8) out of bracket (12).

Loosen screw (25) to remove front brake hose (15) and ABS sensor cable (B) from hose clip (24).



Loosen the screw (16) and slide the front control unit-calliper hose (26) from the bracket (12) on the bottom yoke.



Remove the hoses (8) and (26) from ABS control unit (Sect. G 9.2, [Removing of the ABS control unit](#)).



Warning

While removing the front brake hose, if you damage the hose clip (24) you shall renew it (Sect. E 4, [Removal of the front mudguard](#)).

If hose is not fastened by hose clip (24), it might interfere with tyre under braking and provoke accidents.

Remove the master cylinder (Sect. F 3, [Removal of the front brake master cylinder](#)).

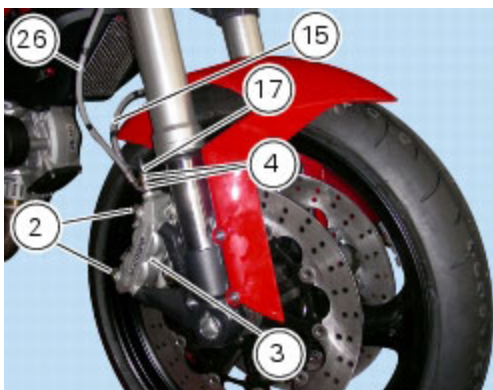
Unscrew the two retaining screws (2) securing the left front brake calliper (10) to the fork leg.

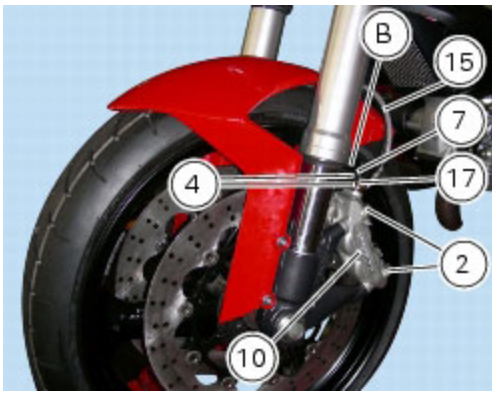
Repeat the operation for the right brake calliper (3).

Free ABS sensor cable (B) from the dust seal (7) on front left-hand brake calliper (10).

Unscrew the banjo bolts (17) from both callipers and recover the sealing washers (4).

Detach the front brake callipers from the hoses (26) and (15).





Note

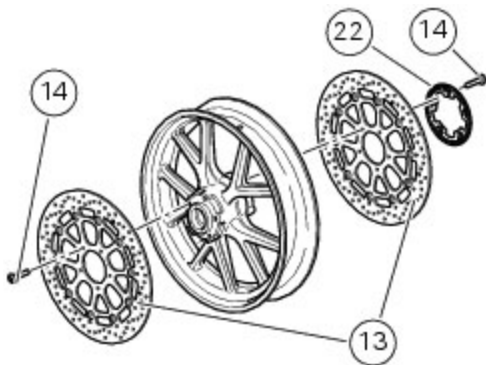
For correct position of the ties fastening the ABS sensor cable (B) to front brake lines (15), please refer to Sect. G 9.2, [Flexible wiring/hoses positioning](#).

Removal of the brake discs

Operation	Section reference
Remove the front wheel	G 1, Removal of the front wheel

The front brake discs consist of an inner carrier, which is mounted to the wheel, and an outer rotor. Both parts must be changed together as a pair.

Undo the fixing screws (14) of the disc to the wheel, remove the disc (13) and, on its left side, slide out the phonic wheel (22).



Overhauling the front brake components



Important

Critical safety components. The manufacturer of the brake calliper advises against servicing the internal components of callipers. Incorrect overhaul of this component could endanger rider safety.

Maintenance operations on these units are limited to replacing the following parts: pads, fasteners and bleed valve assembly.

Refer to the exploded view at the beginning of this section for indications on renewal of the above components.

The brake disc must be clean without any signs of rust, oil, grease or dirt and no deep scoring.

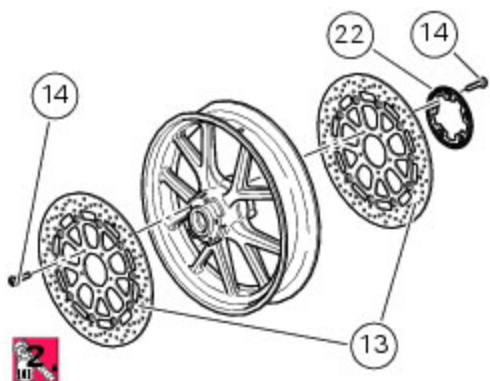
To check brake disc wear, refer to the service limits given in the paragraph "[Hydraulic brakes](#)" (Sect. C 1.1).

Refitting the brake discs

Before refitting the brake disc to the wheel, clean all contact surfaces thoroughly and smear a medium strength threadlocker on the threads of retaining screws (14).

Operating on the left side, fit the phonic wheel (22).

Screw in the screws (14) securing the front brake disc (13) to the front wheel in the sequence: 1-3-5-2-4.

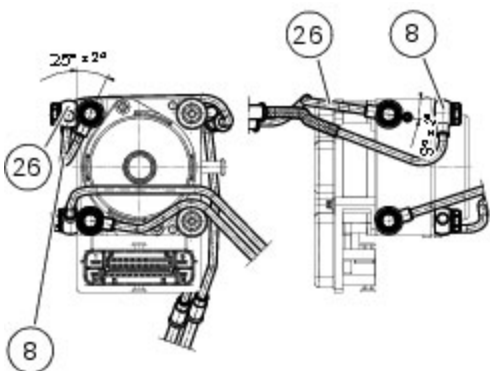
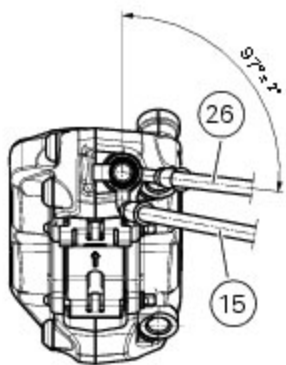
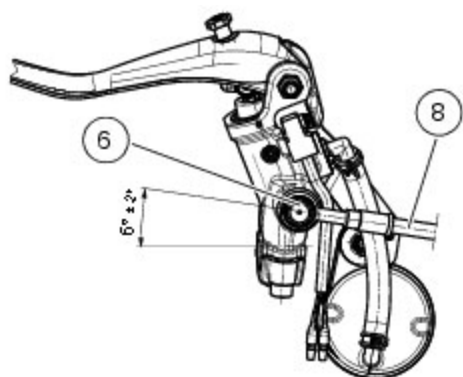


Tighten the screws to the specified torque (Sect. C 3, [Frame torque settings](#)).

Operation	Section reference
Refit the front wheel	G 1, Refitting the front wheel

Refitting the front brake system

When reassembling the system, pay special attention to the position of the end fittings of front master cylinder-control unit hose (8) (onto cylinder and control unit), of control unit-front calliper hose (26) (onto control unit and calliper) and front brake line (15) (on callipers).



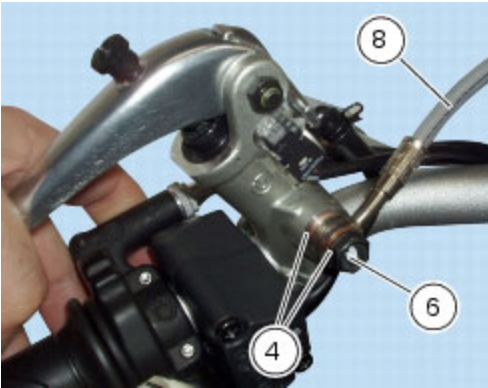


Warning

If incorrectly positioned, the hose can affect brake operation and foul moving parts. Position the hose as shown in the figure.

When reconnecting the brake hose to the calliper, the control unit or master cylinder, make sure to install the sealing washers (4) either side of the hose end fitting.

After orientating the coupling, lock the banjo bolt (6) to the specified torque (Sect. C 3, [Frame torque settings](#)).



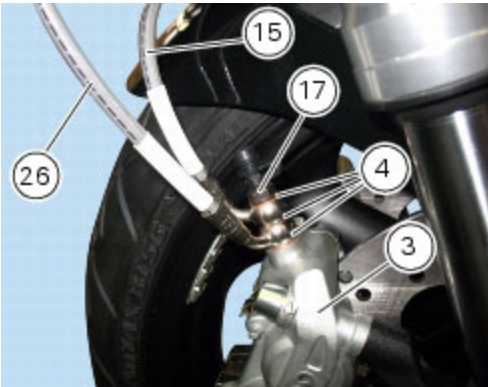
Reconnect the hoses (26) and (15) to the front brake callipers (3) and (10) and tighten the screws (17) to their specified torque (Sect. C 3, [Frame torque settings](#)).



Note

Make sure that internal pipe coupling (26) and the external pipe coupling (15) are installed in the right brake calliper (3).

Duly reassemble the hoses (8) and (26) to the ABS control unit (Sect. G 9.2, [Refitting the ABS control unit](#)).



Fit the RH calliper (3) over the disc.

Apply the recommended grease to the screws (2).

Hand tighten the bolts (2) to secure the callipers to the fork legs.

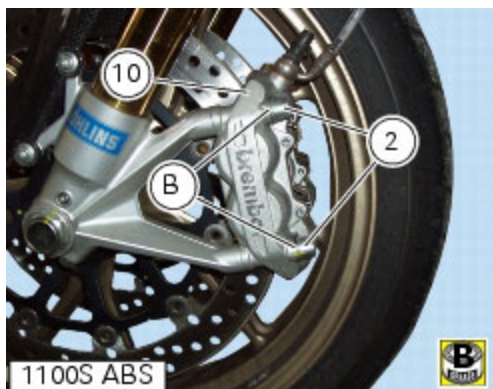
For the 1100S ABS model, fit the washers (B) inserting the screws (2).

Tighten the screws as a first stage to the specified torque (Sect. C 3, [Frame torque settings](#)).

Repeat the operation for the left brake calliper (10).

Operate the front brake lever two or three times until the circuit is pressurised and force the pads against the brake disc.

Hold the lever pulled in towards the grip and tighten the calliper screws (2) to the specified torque (Sect. C 3, [Frame torque settings](#)).

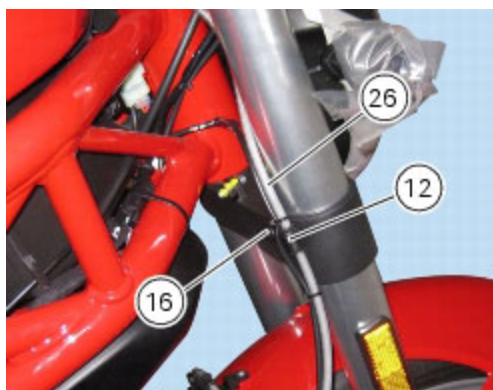
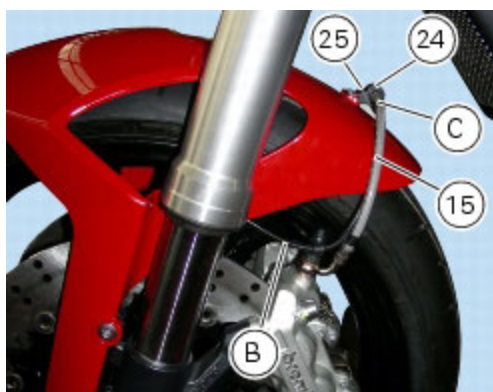


Insert brake rubber block (C) into grommet (24) and ensure it is centred.
 Fold hose clip (24) and fasten it by starting the screw (25) in its thread.
 Insert hose (26) in bracket (12) and tighten screw (16) to the specified torque (Sect. C 3, [Frame torque settings](#)).



Note

For correct position of the ties fastening the ABS sensor cable (B) to front brake lines (15), please refer to Sect. G 9.2, [Flexible wiring/hoses positioning](#).



Filling the braking system

D 4, [Filling the brake system with fluid](#)